

CLEAR-Papaya

Full Execution Roadmap: Prototype to Journal Submission

Generated 27 August 2026 • 16-week plan • 72 tasks • 10 gates

72	10	16	6
Total Tasks	Go/No-Go Gates	Weeks to Submit	Phases

How to use this document: Print it. Every task has a checkbox. Work top to bottom. **RED "GATE" items** are decision points — if a gate fails, STOP and re-evaluate before spending more time. **GOLD "Deliverable" items** are concrete outputs that prove the task is done. Claude can generate any script, template, or draft marked with **[Claude can help]** — just ask.

CRITICAL PRINCIPLE: Week 2 er gate (leakage rate) pass na korle architecture-e invest koro na. Oi ek ta number-i paper-er headline result. Baki 14 week er sob kichu oi number-er upor depend kore.

PHASE 1 — DATA ACQUISITION & LEAKAGE AUDIT

This phase proves the paper's central claim: that published papaya leaf disease accuracy numbers are inflated by near-duplicate leakage. If it works, the paper writes itself. If it doesn't, you learn in 2 weeks instead of 16.

Week 1 — Dataset Collection & Verification

1.1 — Search and list ALL public papaya leaf disease datasets

Google Scholar, Kaggle, Mendeley Data, Roboflow Universe, GitHub, UCI ML Repository, Zenodo search korbē. Query: *papaya leaf disease dataset*. Minimum 3 ta source dorkar.

Deliverable: *Spreadsheet: dataset_name, URL, DOI, image_count, classes, licence, download_tested*

1.2 — Download every dataset, verify file integrity

Protita dataset download koro. ZIP extract koro. Image count match verify koro published description er shathe. Kono corrupt image thakle note koro. **[Claude can help]** — integrity check script.

Deliverable: *data/ folder with all sources, README per source*

1.3 — Verify licence compatibility

Protita source er licence page kholbe. Screenshot nibe. CC BY, CC BY-SA, CC0, Apache 2.0 = OK. NC (non-commercial) = problematic for some venues. Licence pawa na gele = use kora jabe na.

Deliverable: *licence_verification.md with screenshots*

GATE: Kono dataset licence-free hole, seta exclude koro. DOI nei = cite kora jabe na = exclude.

1.4 — Build unified data loader + label mapping

Sob source er class names align koro (e.g. 'PRSV' vs 'Ring Spot' vs 'Mosaic' = same class). Ekta master CSV banao: image_path, source_id, original_label, unified_label. **[Claude can help]** — loader script.

Deliverable: *master_manifest.csv + load_dataset.py*

1.5 — Compute basic dataset statistics

Per-source per-class image count, resolution distribution (min/max/median), aspect ratio distribution, file size distribution. Table 5 er data eta theke ashbe.

Deliverable: *dataset_stats.json + Table 5 numbers filled in LaTeX*

Week 2 — Leakage Audit (THE CRITICAL GATE)

This is the most important week of the entire project. If $\rho > 0.3$, your paper has its headline. If $\rho < 0.05$, you need a different angle.

2.1 — Compute MD5 + perceptual hash (pHash) for every image

MD5 = exact duplicates. pHash (64-bit DCT) = near-duplicates (resized, re-compressed, slightly cropped). Hamming distance threshold $\theta_p = 5$. **[Claude can help]** — full script ready to generate.

Deliverable: *hashes.csv: image_path, md5, phash_hex*

2.2 — Extract DINOv2 embeddings for every image

DINOv2 ViT-S/14 (Facebook, torch.hub). GPU lagbe — free Colab-e hobe. CLIP ViT-B/32 as robustness check. Embeddings save koro .npy file-e. **[Claude can help]** — embedding extraction script for Colab.

Deliverable: *embeddings_dinov2.npy + embeddings_clip.npy*

2.3 — Build similarity graph + union-find grouping

pHash pairs ($d_H \leq 5$) + embedding cosine pairs ($\cos \geq \theta_e$) ke edge hisebe treat koro. Union-find diye connected components ber koro = leaf-instance groups. θ_e calibrate korte 100-200 ta image pair manually label koro (same leaf / different leaf). **[Claude can help]** — union-find + calibration script.

Deliverable: *groups.csv: image_path, group_id, group_size | θ_e calibration.json*

2.4 — Compute leakage rate ρ for a standard random split

70/15/15 random split koro 10 bar. Protitar ρ compute koro (Eq. 2 of paper). Mean +/- sd report koro. **[Claude can help]** — split + ρ calculation script.

Deliverable: *leakage_report.json: mean_rho, std_rho, per_split_rho*

GATE: $\rho > 0.3$ = PROCEED (strong result). $\rho 0.1-0.3$ = moderate, adjust framing. $\rho < 0.05$ = STOP, rethink angle.

2.5 — Visualize duplicate clusters

Shobcheye boro 10 ta group er sob images ekta grid-e plot koro. Eta Figure 2 er raw material — reviewer ke dekhabe 'same leaf, different photo'. **[Claude can help]** — grid visualization script.

Deliverable: *duplicate_clusters_top10.png (for paper Fig. 2)*

❏ 2.6 — Create group-stratified 5-fold split (S2)

GroupKFold use koro, group = group_id. Class balance preserve koro (StratifiedGroupKFold). Verify: $\rho = 0$ for every fold. Save fold assignments. **[Claude can help]** — split script.

Deliverable: *splits_S2.csv: image_path, fold (0-4) | verified $\rho=0$ for each fold*

PHASE 2 — FIELD DATA COLLECTION & ANNOTATION

Week 3-4 — Field Collection (D6: NU-PapayaField)

❑ 3.1 — Identify 3+ collection sites across different districts

Home orchards, nurseries, commercial plots — variety dorkar. Minimum 3 ta separate location. Landowner permission nao (written or verbal, documented).

Deliverable: *sites.csv: site_id, district, gps_coarsened, type, permission_status*

❑ 3.2 — Prepare 2-3 different smartphones for capture

Different price tier er phone use koro (e.g. budget Xiaomi, mid-range Samsung, flagship). Device identity record koro — eta shift variable, noise na.

Deliverable: *devices.csv: device_id, model, camera_mp, OS_version*

❑ 3.3 — Photograph leaves IN SITU (natural light only)

NO white background. NO studio setup. Real field conditions. Per leaf: 3 captures at different angles. CRITICAL: same leaf_id for all 3. Target: minimum 50 images per class = 300+ total images. **[Claude can help]** — metadata recording template/app.

Deliverable: *raw_images/ + metadata.csv (schema from Appendix D of paper)*

❑ 3.4 — Record metadata per capture

image_id, leaf_id, plant_id, plot_id, district, date, time, device, illumination_class (sun/overcast/shade/dusk), phenostage. CSV template use koro. **[Claude can help]** — CSV template.

Deliverable: *d6_metadata_complete.csv*

❑ 3.5 — Get 2 independent annotators to label

Annotator 1 = tumi (CSE background). Annotator 2 = plant pathology person (MUST find). Independently label every image into c1-c8. Record confidence (1-5). Disagreements resolved by 3rd person. **[Claude can help]** — annotation agreement calculation.

Deliverable: *annotations.csv with annotator_id columns | Cohen's kappa computed*

GATE: GATE: $\text{kappa} < 0.6$ = labels unreliable = reviewer will reject. Retrain annotators or simplify taxonomy.

❑ 3.6 — Create expert lesion masks (subset of 30+ per class)

Stratified subset select koro. Polygon annotation tool use koro (LabelMe, CVAT, or Roboflow). Expert traces the lesion boundary. Eta XAI evaluation-er jonno (Section 5.8). Plant pathology annotator-ke bolte hobe.

Deliverable: *masks/ folder with polygon JSON/PNG per image*

❑ 3.7 — Validate D6 with the leakage audit

D6 er upor-o leakage audit chalao (same script as 2.1-2.3). D6 er rho should be 0 by construction (because leaf_id is known). Report inferred-group vs ground-truth-leaf_id precision/recall.

Deliverable: *d6_audit_report.json: group_precision, group_recall*

PHASE 3 — BASELINE EXPERIMENTS & THE HEADLINE RESULT

Week 5-6 — Train Baselines Under Both Splits

❑ 5.1 — Set up training environment

Google Colab Pro recommended. Record: GPU type, CUDA version, PyTorch version, timm version. Exact versions — 'a GPU' is not a specification. Create requirements.txt / environment.yml. **[Claude can help]** — training boilerplate.

Deliverable: *env_spec.txt with exact versions | requirements.txt*

❑ 5.2 — Train 4 key models under S1 (random split, leaky)

ResNet-50, EfficientNet-B0, MobileViT-S, MobileNetV3-Large. Identical hyperparameters (Table 8 of paper). 5 seeds x single split = 5 runs per model. Record: accuracy, macro-F1, training time, best epoch. **[Claude can help]** — training script.

Deliverable: *results_S1.csv with per-seed metrics*

❑ 5.3 — Train SAME 4 models under S2 (group split, clean)

Same hyperparameters. Same seeds. 5 seeds x 5 folds = 25 runs per model. This is compute-heavy but non-negotiable. Record identical metrics.

Deliverable: *results_S2.csv with per-seed-per-fold metrics*

❑ 5.4 — Compute Delta = S1_accuracy - S2_accuracy per model

Paired comparison. Report mean delta, sd, p-value (5x2cv paired t-test). This is Table 12 of the paper — THE headline result. **[Claude can help]** — statistical test script.

Deliverable: *Table 12 of paper completely filled*

GATE: GATE: If Delta > 5pp for most models = STRONG paper. If Delta < 2pp = the correction is real but small, reframe as 'the protocol matters even when the gap is modest'.

❑ 5.5 — Clever-Hans background probe (3-way ablation)

Segment leaves with SAM (segment-anything). Create: full / leaf-only / bg-only variants. Evaluate SAME trained models on all three. Report Acc(bg-only) vs chance (1/K). If bg-only >> chance = dataset has background confounding. **[Claude can help]** — SAM segmentation + ablation script.

Deliverable: *Table 13 of paper filled | segmentation_iou.json*

GATE: GATE: If bg-only accuracy ~ chance = dataset is clean (good, but less dramatic). If bg-only >> chance = strong finding, report prominently.

Week 7 — Full Baseline Sweep

❑ 7.1 — Train ALL baseline models from Table 11

Add: DenseNet-121, ConvNeXt-Tiny, ViT-B/16, DeiT-S, Swin-T, ResNet-50+SE, ResNet-50+CBAM, GLCM+LBP-SVM, GLCM+LBP-RF. S2 split only (group-stratified). 5 seeds x 5 folds.

Deliverable: *results_all_baselines_S2.csv | Table 11 of paper filled*

❑ 7.2 — Friedman test + Nemenyi post-hoc + critical difference diagram

Multi-model comparison. Holm-Bonferroni correction. Critical difference diagram generate koro (scikit-posthocs or Orange). **[Claude can help]** — statistical comparison script.

Deliverable: *critical_difference_diagram.pdf*

PHASE 4 — PAPAYAFORMER ARCHITECTURE & TRAINING

Week 8 — Implement PapayaFormer

8.1 — Implement MSLA module

Multi-Scale Lesion Attention: 3 dilated conv branches + spatial sigmoid + channel SE. Learnable gamma initialized at 0. Unit test: output shape matches input shape; at init, output == input (identity). **[Claude can help]** — PyTorch module code.

Deliverable: *msla.py + test_msla.py (passing)*

8.2 — Implement Evidential Dirichlet head

Softplus activation $\rightarrow \alpha = e + 1 \rightarrow$ Dirichlet. EDL loss with KL annealing. Unit test: u in $(0,1]$, p sums to 1, gradients flow. **[Claude can help]** — EDL head + loss code.

Deliverable: *evidential_head.py + test_edl.py (passing)*

8.3 — Assemble full PapayaFormer backbone

MobileViT-S backbone from timm. Insert MSLA at stages 2, 3, 4. Replace classifier head with evidential head. Load ImageNet weights for non-MSLA parts. Sanity check: overfit on 10 images (loss $\rightarrow 0$ within 50 epochs).

Deliverable: *papayaformer.py | overfit_sanity.log*

GATE: GATE: If overfitting sanity check fails = bug in architecture. Debug before proceeding.

8.4 — Implement composite loss (Eq. 16)

$L_{EDL} + \lambda_1 * L_{CB} + \lambda_2 * L_{con} + \lambda_3 * L_{attn}$. KL annealing over first T_{anneal} epochs. Class-balanced focal weights from class frequencies. **[Claude can help]** — loss implementation.

Deliverable: *losses.py + test_losses.py*

Week 9 — Train PapayaFormer + Ablations

9.1 — Train PapayaFormer on S2, 5 seeds x 5 folds

Same protocol as baselines. AdamW, cosine LR, EMA. Early stop on val macro-F1. Save best checkpoint per fold. Record all metrics from Section 5.2.

Deliverable: *results_papayaformer_S2.csv | checkpoints/*

9.2 — Run component ablation (Table 16)

Remove one component at a time: -MSLA, -evidential head (softmax instead), -class-balanced focal, -consistency loss, -attention reg, -transformer stages (pure conv), -ImageNet init (scratch), +MixUp, +input 384. Each variant: 5 seeds x 5 folds.

Deliverable: *Table 16 completely filled | ablation_results.csv*

9.3 — Hyperparameter sensitivity grid

Vary λ_1 , λ_2 , λ_3 , T_{anneal} , MSLA dilations, reduction r . Report sensitivity to each (Appendix B of paper). Document failure modes (KL collapse, mask collapse).

Deliverable: *Appendix B tables filled | failure_mode_curves.png*

PHASE 5 — ADVANCED EVALUATIONS (RQ4-RQ10)

Week 10 — Distribution Shift + Low-Data

10.1 — Cross-source evaluation (S3)

Train on D1+D2, test on D3+D6. No tuning on test sources. Report per-model macro-F1 drop from in-distribution.

Deliverable: *Table 14 cross-source columns filled*

10.2 — Corruption robustness suite (S4)

15 corruptions x 5 severities on S2 test set. Use imagecorruptions library. Corruptions: gaussian_noise, shot_noise, impulse_noise, defocus_blur, glass_blur, motion_blur, zoom_blur, snow, frost, fog, brightness, contrast, elastic, pixelate, jpeg. **[Claude can help]** — corruption script.

Deliverable: *corruption_heatmap.png (Figure slot 3) | rCE per model*

10.3 — Low-data regime (S6, RQ10)

Subsample S2 training at {1, 5, 10, 25, 50, 100}% by GROUP (not by image). Train PapayaFormer + 2 baselines at each fraction. Plot learning curve with 95% CI. **[Claude can help]** — subsampling + plotting script.

Deliverable: *Figure slot 6 (data efficiency curve)*

Week 11 — Open-Set + Calibration

11.1 — Open-set evaluation (S5, RQ5)

Add c7 (herbicide injury) + c8 (non-papaya from PlantVillage) to test only. Use uncertainty u as rejection score. Compare: max-softmax, energy score, MC-dropout (20 passes), deep ensemble (5). Report AUROC, AUPR-Out, FPR@95.

Deliverable: *Table 15 completely filled*

11.2 — Calibration analysis (RQ6)

Reliability diagram (15 equal-mass bins). ECE, MCE, Brier score. Apply temperature scaling as comparator. Risk-coverage curves + AURC. Accuracy at coverage {100, 90, 80, 70}%. **[Claude can help]** — calibration script.

Deliverable: *Table 17 filled | reliability_diagram.pdf | risk_coverage.pdf*

Week 12 — Explainability Evaluation

12.1 — Generate attribution maps (4 methods + MSLA intrinsic)

Grad-CAM, Grad-CAM++, Score-CAM, SHAP (partition explainer). Plus MSLA spatial mask (intrinsic, free from the model). Generate for entire XAI evaluation subset.

Deliverable: *attributions/ folder with maps per method per image*

12.2 — Quantitative XAI metrics

Deletion AUC (lower=better), Insertion AUC (higher=better) at 1% steps. Pointing game against expert lesion masks. Attribution mass inside leaf (Omega). Sanity checks: model randomization + label randomization. **[Claude can help]** — XAI evaluation script.

Deliverable: *Table 18 filled | sanity_check_results.json*

12.3 — Qualitative XAI figure WITH failure cases

Grid: 2 examples per class x {input, expert mask, Grad-CAM++, MSLA mask}. MUST include at least 2 FAILURE cases. Cherry-picked successes only = reviewer rejection.

Deliverable: *Figure slot 5 (XAI qualitative grid)*

Week 13 — Edge Deployment Measurement

13.1 — Export to ONNX + TFLite

PapayaFormer + MobileNetV3 + EfficientNet-B0 + ResNet-50. FP32, FP16, INT8 (post-training quantization, 500-image calibration from training data). **[Claude can help]** — export script.

Deliverable: *models/ with .onnx and .tflite per variant*

13.2 — Measure latency on Android device

Mid-range Android phone. TFLite benchmark tool. 200 runs after 20 warmup. 1-thread and 4-thread. Report median and p95. Peak RSS. Battery-delta energy measurement over 1000 inferences.

Deliverable: *android_benchmark.csv | Table 19 Android columns filled*

❏ 13.3 — Measure latency on Raspberry Pi 4

TFLite runtime on RPi4 4GB. Same protocol: 200 runs, median/p95. USB inline power meter for energy per inference. If no RPi4 available, state it as a limitation and measure on another ARM device.

Deliverable: *rpi4_benchmark.csv | Table 19 RPi4 columns filled*

❏ 13.4 — Per-class accuracy of quantized variants

INT8 accuracy per class vs FP32 per class on SAME S2 test set. Quantization typically destroys minority class first — report this per-class delta. Appendix C INT8 column.

Deliverable: *quantization_perclass.csv | Appendix C INT8 column filled*

PHASE 6 — WRITING, ARTEFACT RELEASE & SUBMISSION

Week 14 — Statistical Finalization + All Figures

❑ 14.1 — Compute ALL bootstrap confidence intervals

BCa bootstrap, B=10000, resampled at GROUP level (not image level). 95% CI for every primary metric in every table. [Claude can help] — bootstrap CI script.

Deliverable: Every table CI column filled

❑ 14.2 — Run ALL significance tests

5x2cv paired t-test for pairwise comparisons. McNemar for paired errors. Friedman + Nemenyi for multi-model. Holm-Bonferroni correction within each family. Cliff's delta effect size alongside every p-value.

Deliverable: Every table p-value and delta column filled

❑ 14.3 — Generate ALL camera-ready figures

Confusion matrix (normalized), t-SNE/UMAP embedding plot, corruption heatmap, data efficiency curve, reliability diagram, risk-coverage curve, duplicate cluster grid, XAI grid. [Claude can help] — matplotlib figure scripts.

Deliverable: figures/ with all PDFs | All figure slots in LaTeX filled

❑ 14.4 — Fill EVERY remaining TBD in the LaTeX manuscript

Search for every '--' (TBD marker) in the LaTeX source. Replace with actual numbers. If a number can't be filled, delete the row/claim. Never estimate or invent a result.

Deliverable: grep -c 'TBD' paper.tex returns 0

GATE: GATE: Any remaining TBD = not submission ready. Do not submit.

Week 15 — Write Discussion/Conclusion + Release Artefacts

❑ 15.1 — Write Discussion section

Interpret leakage result (tone: retiring a convention, not accusing). Background probe implications. Deployment operational statement. Agronomic cost-sensitive analysis. SOTA positioning. [Claude can help] — draft Discussion from your results.

Deliverable: Discussion section written in LaTeX (Section 7)

❑ 15.2 — Write Conclusion

Mirror abstract. No new claims. Name the state-of-the-art number under corrected protocol. 5 concrete future work items.

Deliverable: Conclusion section written (Section 9)

❑ 15.3 — Fill Related Work table (Table 2) from screening

Verify every prior work entry: exact year, exact accuracy, exact dataset, exact split procedure (quote verbatim from their Methods). NEVER copy from secondary sources. Report screening counts (Appendix E).

Deliverable: Table 2 + Appendix E completely filled

GATE: GATE: Any citation you have not read at least abstract + methods = remove it. Fabricated citations end careers.

❑ 15.4 — Deposit data + code on Zenodo

Create Zenodo deposit: D6 dataset, split manifests (with hashes and group_ids), expert lesion masks, all configs, all evaluation scripts, checkpoints. CC BY 4.0 for data, MIT for code. Get DOI. Put DOI in paper.

Deliverable: Zenodo DOI minted and live

❑ 15.5 — Create GitHub repo with reproduction script

README with 'make reproduce' or single-command instruction. Environment lockfile. Tagged release matching manuscript version.

Deliverable: GitHub URL in paper | make reproduce works end-to-end

❑ 15.6 — Verify Appendix A (Reproducibility Checklist)

Go through all 23 items. Mark status honestly. Every 'TBD' must become checkmark or 'N/A (reason)'. This checklist IS your submission readiness test.

Deliverable: All 23 checklist items resolved

Week 16 — Final Review & Submit

❑ 16.1 — Internal review against Appendix F (Reviewer Objections)

Read each of 12 anticipated objections. For each, find the EXACT paragraph/table in the manuscript that answers it. If you can't find it = add it.

Deliverable: objection_crossref.md mapping each objection to manuscript location

❑ 16.2 — Add Generative AI disclosure

Required by Elsevier, IEEE, Springer, Nature. State tool name and usage. Omission is now a desk-reject trigger at most venues.

Deliverable: Declaration of generative AI use in Declarations section

❑ 16.3 — Language + formatting pass

Read aloud. Fix awkward phrasing. Check: consistent terminology (same word for same concept), figure/table numbering sequential, all cross-references resolve, bibliography complete. **[Claude can help]** — proofreading pass.

Deliverable: No LaTeX warnings for undefined references

❑ 16.4 — Format for target venue

Download venue's LaTeX template. Reformat (if not already compatible). Check: page limit, reference format, figure resolution requirements, supplementary material policy. **[Claude can help]** — template conversion.

Deliverable: Manuscript formatted in venue template

❑ 16.5 — SUBMIT

Cover letter. Suggested reviewers (3-5 names in the field, with justification). Excluded reviewers (if any). Upload all files. DONE. **[Claude can help]** — cover letter draft.

Deliverable: Submission confirmation email received

VENUE TARGET LIST

Verify current scope, IF, APC, and template at the journal website before choosing. These change annually.

Venue	Type	Fit	Priority
Computers and Electronics in Agriculture (Elsevier)	Q1, hybrid	Best fit: ag ML + field validation, expects agronomic depth	1st choice
Smart Agricultural Technology (Elsevier)	Q1/Q2, OA	Deployment + edge focus matches RQ9 material	2nd choice
Frontiers in Plant Science	Q1, OA	Precedent (Mohanty 2016), correction narrative fits	3rd choice
Plant Methods (BMC)	Q1, OA	Strong fit for protocol/benchmark contribution	Alternate
Scientific Reports (Nature)	Q1, OA	Methodology-correction framing travels well	Alternate
IEEE Access	Q1/Q2, OA	Fast turnaround, tolerant of broad benchmark work	Fallback
Artificial Intelligence in Agriculture	Q1/Q2, OA	Reasonable, growing reputation	Fallback

SUBMISSION NON-NEGOTIABLES

1. Every TBD replaced by a number from YOUR experiments, or the row deleted. Never estimate.
2. Every dagger-marked item verified at the PRIMARY source. Never cite a DOI you haven't opened.
3. Every citation in the bibliography read by YOU (at least abstract + methods). Fabricated citations end careers.
4. Claim about prior work stated as 'the reported procedure does not exclude leakage', NEVER as 'they leaked'.
5. Data and code deposited, DOI minted, and links LIVE before submission — not 'available upon request'.
6. Generative AI disclosure included. Omission = desk reject at Elsevier/IEEE/Springer/Nature in 2026.
7. No test-set tuning. Threshold tau fixed on validation fold. State this explicitly.
8. Failure cases shown in qualitative figures. All-success cherry-picking = reviewer rejection.

WHAT CLAUDE CAN GENERATE FOR YOU

Just ask for any of these by name. They are ready to be generated on demand:

Script / Deliverable	What it does
Leakage audit pipeline	pHash + DINOv2 + union-find + rho calculation. Full Python script, runs on Colab.
Training boilerplate	PyTorch training loop with timm, EMA, cosine LR, wandb logging, early stopping.
PapayaFormer architecture	MSLA module + evidential head + composite loss. PyTorch code + unit tests.
SAM segmentation script	Clever-Hans background probe: segment, ablate, evaluate.
Corruption suite	15-corruption x 5-severity evaluation script using imagecorruptions.
XAI evaluation	Deletion/insertion AUC, pointing game, sanity checks. Full pipeline.
Edge export + benchmark	ONNX/TFLite export + INT8 quantization + benchmark harness.
Statistical analysis	Bootstrap CIs, 5x2cv paired t-test, Friedman+Nemenyi, Cliff's delta. All in one script.
All figures	matplotlib scripts for every figure slot in the paper.
Field metadata CSV template	Ready-to-fill template matching Appendix D schema.
Cover letter	Venue-specific cover letter draft.
Discussion/Conclusion draft	From your actual results, in the paper's established voice.

LaTeX finalization	Proofread, fix references, format for target venue template.
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